

04 — RAPID FIRE (100 Q&A)

Cover the question, say the answer, then check. Morning-before-exam drill.

Chapter 1 — Basics of SQA

1. Two major types of software? — Generic (Buy) and Customized (Build).
2. What does COTS stand for? — Commercial off-the-shelf.
3. "Bespoke" software is which type? — Customized (Build).
4. Software is composed of what? — Programs, procedures, documentation, and data.
5. Software quality per ISO/IEC 9126? — Totality of functionality and features that satisfy stated or implied needs.
6. Software quality per IEEE Std 610? — Degree to which a system/component/process meets specified requirements and user needs.
7. Three main project challenges? — Time, Cost, Scope.
8. What was Therac-25? — A radiation-therapy machine (1986) that overdosed patients due to a software bug.
9. Which company built Therac-25? — AECL (Atomic Energy of Canada Ltd).
10. When did Ariane 5 fail, and how long after launch? — 4 June 1996, 37 seconds after launch.
11. Why is Ariane 5 famous? — The most expensive computer bug in history (control-software fault).
12. Definition of software testing? — Executing a system under specified conditions to find defects and verify requirements.
13. Two categories of quality assessment? — Static (manual) analysis and Dynamic (automated) analysis.
14. Which analysis executes the program? — Dynamic analysis.
15. Three examples of static analysis? — Code review, inspection, algorithm analysis.
16. SQA is what kind of activity? — An umbrella activity applied throughout the software process.
17. Does SQA execute the program? — No.
18. Does SQC execute the program? — Yes.
19. Who is responsible in SQA vs SQC? — SQA = all team members; SQC = only the testing team.
20. SQA and SQC map to which of Verification/Validation? — SQA = Verification; SQC = Validation.
21. Define Error / Fault / Failure / Defect (one line each)? — Error = human action producing wrong result; Fault = incorrect step/data in the program; Failure = system can't perform its function; Defect = deviation from spec.
22. Three stages of SqE? — Planning, Execution, Decision.

Chapter 2 — Software Quality

23. External view sees software as? — A black box.
24. Internal view sees software as? — A white / clear box.
25. Who proposed the five views of quality? — Kitchenham and Pfleeger.
26. Name the five views of quality. — Mystical, User, Manufacturing, Product, Value-based.
27. Which view underlies CMM and ISO 9001? — Manufacturing view.
28. Which view says quality is recognized but not defined? — Mystical view.
29. The value-based view merges which two concepts? — Excellence (quality) and worth (value).
30. Product view claim? — Good internal properties lead to good external properties.

- 31.** Three reasons to measure quality? — Baseline; quality improvement based on cost; know present level for planning.
- 32.** What is a quality factor? — A behavioural characteristic of a system.
- 33.** Examples of quality factors? — Correctness, reliability, efficiency, performance.
- 34.** What is a quality criterion? — An attribute of a quality factor, related to development.
- 35.** Examples of quality criteria? — Modularity, testability, maintainability, reusability.
- 36.** Whose model gives factors and criteria? — McCall's.
- 37.** Definition of reliability? — Probability of executing without failure for a specific period of time.
- 38.** Most influential quality framework? — ISO-9126.
- 39.** How many ISO-9126 top-level characteristics, and what are they? — Six: Functionality, Reliability, Usability, Efficiency, Maintainability, Portability.
- 40.** Is security an ISO-9126 top-level characteristic? — No.

Chapter 3 — Maturity Models

- 41.** What is a software process? — A set of activities executed to develop products.
- 42.** Three benefits of a defined process? — Repeatable, evaluable (via metrics), improvable.
- 43.** Three reasons to improve a process? — Quality, Lead Time, Cost.
- 44.** What does CMM evaluate? — Software development processes.
- 45.** What does TMM evaluate? — A testing process.
- 46.** What does TPI do? — Improves the testing process.
- 47.** How many CMM levels? — 5.
- 48.** Name the five CMM levels. — Initial, Repeatable, Defined, Managed, Optimizing.
- 49.** CMM Level 1? — Initial.
- 50.** CMM Level 4? — Managed.
- 51.** CMM Level 5? — Optimizing.
- 52.** Which CMM level uses data collection and analysis? — Level 4 (Managed).
- 53.** How many TMM levels? — 5.
- 54.** Who pioneered TMM? — Ilene Burnstein.
- 55.** Name the five TMM levels. — Initial, Phase Definition, Integration, Management and Measurement, Optimization.
- 56.** How many steps to improve a test process (TPI)? — 4.

Chapter 4 — Software Quality Assurance

- 57.** Three generic QA categories? — Defect Prevention, Defect Reduction, Defect Containment.
- 58.** Two ways to do defect prevention? — Error source removal and error blocking.
- 59.** Error source removal is done via? — Education and training (people-based).
- 60.** Most common static defect-reduction technique? — Inspection.
- 61.** Two forms of inspection formality? — Informal walkthroughs/reviews and formal inspections.
- 62.** Two ways to do defect containment? — Software fault-tolerance and safety assurance.
- 63.** What is NVP? — N-Version Programming (a fault-tolerance technique).
- 64.** Recovery in fault tolerance means? — Rollback and redo.

- 65. Define safety / accident / hazard. — Safety = accident-free; Accident = failure with severe consequences; Hazard = pre-condition to an accident.
- 66. Is defect prevention 100% effective? — No.
- 67. Name the four development process variations. — Waterfall, Iterative, Spiral, Agile (plus Maintenance).
- 68. In the DC→V&V mapping, unit testing maps to? — Verification.

Chapter 5 — Software Quality Engineering

- 69. What does SQE minimize while meeting quality? — Cost and project risks.
- 70. Three phases of the generic testing process? — Pre-QA, In-QA, Post-QA.
- 71. When are most key testing decisions made? — Pre-QA / planning.
- 72. Test plan is which level? — High-level.
- 73. Test case is which level? — Low-level.
- 74. What is a test suite? — A collection of test cases run until stopping criteria (macro-level).
- 75. Reusing earlier test cases is called? — Regression testing.
- 76. Does "Post-QA" mean after QA finishes? — No; it runs parallel to QA.
- 77. Three testing team models? — Vertical, Horizontal, Mixed.
- 78. The horizontal model does what? — One kind of testing for many different products.

Chapter 6 — Software Testing

- 79. How many principles of testing? — 7.
- 80. Principle 1? — Testing shows presence of defects, not their absence.
- 81. Principle 2? — Exhaustive testing is impossible.
- 82. Principle 3? — Early testing (defects are cheaper to fix early).
- 83. Principle 4? — Defect clustering.
- 84. Principle 5? — Pesticide paradox.
- 85. Principle 6? — Testing is context dependent.
- 86. Principle 7? — Absence-of-errors fallacy.
- 87. Four testing levels? — Unit, Integration, System, Acceptance.
- 88. Who does unit testing? — Developer.
- 89. Who does integration and system testing? — Tester.
- 90. Who does acceptance testing? — End-user.
- 91. Does regression testing design new tests? — No; it selects, prioritizes, and executes existing ones.
- 92. White-box testing is also called? — Structural / implementation-based testing.
- 93. Black-box testing is also called? — Functional / specification-based testing.
- 94. Two things white-box examines? — Control flow and data flow.
- 95. Who performs white-box testing? — Developers.
- 96. Who performs black-box testing? — Professional / third-party testers (IV&V).
- 97. What is gray-box testing? — Mixed black-box and white-box (middle levels).
- 98. Is "no more defects found" a valid stopping criterion? — No.
- 99. Two types of exit criteria? — Resource-based and quality-based.
- 100. Can automation replace manual testing? — No.